

parts are equal, so $b = 5$. We have gotten another letter of the alphabet, and now we need to simplify $3/6$ to progress further. If you can't tell immediately that $3/6$ is equal to $1/2$, you could follow the same process we did for $6/33$: say that $3/6$ expressed as a reciprocal is $1/(6/3)$, and we find that $6/3$ is 2 remainder 0. The algorithm we're following is meant to complete when we have a remainder of 0, so we will realize that we've finished the process, and we can write our full continued fraction as in [Listing 5-3](#).

$$\frac{105}{33} = 3 + \frac{1}{5 + \frac{1}{2}}$$

[*Listing 5-3: A continued fraction for 105/33*](#)

If this process of repeatedly dividing two integers to get a quotient and a remainder felt familiar to you, it should have. In fact, it's the same process we followed in Euclid's algorithm in Chapter 2! We follow the same steps but record different answers: for Euclid's algorithm, we recorded the final nonzero remainder as the final answer, and in the continued fraction generation algorithm, we recorded every quotient (every letter of the alphabet) along the way. As happens so often in math, we have found an unexpected connection—in this case, between the generation of a continued fraction and the discovery of a greatest common divisor.

We can implement this continued fraction generation algorithm in Python as follows.

We'll assume that we're starting with a fraction of the form x/y . First, we decide which of x and y is bigger and which is smaller:

```
x = 105
y = 33
big = max(x,y)
small = min(x,y)
```

Next, we'll take the quotient of the bigger divided by the smaller of the two, just as we did with $105/33$. When we found that the result was 3,